

Vigneshwaran C. J.

AI / ML Engineer and Full-Stack Developer

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PROFILE

Full-stack developer and AI / ML engineer with an M.Tech in Big Data Biology. Work spans enterprise applications, WhatsApp automation, real-time networking, and computational biology tools.

EXPERIENCE

Junior Software Developer | Ceiyone Tech Works Pvt. Ltd.

Dec 2025 - Present

Designing, developing, and maintaining enterprise-grade applications and AI-powered platforms - including AIORA, a multi-tenant WhatsApp business platform of 14 services (bookings, storefronts, n8n-driven AI conversations) running on a shared Hostinger VPS under systemd/PM2 behind nginx, with Render- and Railway-hosted front ends. Work spans backend APIs, WhatsApp automation, cloud deployments, AI/LLM integration, and internal monitoring. Also mentoring 2 engineering interns on the AIORA platform.

AI Engineer Intern | Ceiyone Tech Works Pvt. Ltd.

Oct 2025 - Dec 2025

Built AI-powered automation solutions, OCR-based document systems, LinkedIn automation tools, and multi-agent LLM systems. Integrated AI services, external APIs, and workflow automation platforms.

Research Intern | Manipal Institute of Technology

Jun 2024 - Jul 2024

Developed ML-based QSPR models to predict the solubility of Posaconazole in Self-Emulsifying Drug Delivery Systems (SEDDS). Used RDKit for molecular descriptor generation (50+ descriptors), VIF analysis for feature selection (10 key descriptors: ELUMO, EHOMO, MaxQ-, MaxQ+, MinPartialCharge, MaxPartialCharge, MinAbsPartialCharge, FpDensityMorgan2, PEOE_VSA1, HallKierAlpha). Evaluated Linear Regression, Random Forest, PLS, KNN, and SVM models.

Tutor | Kamakshi Tuition Centre

Jan 2023 - Jun 2023

Taught mathematics to 9th grade students and Python programming to 12th grade students at Kamakshi Tuition Centre, Mugalivakkam Arasa Maram.

SELECTED PROJECTS

Syncly

Developer-to-developer networking platform built solo end to end. Members pick a mentorship role (mentor, learner, both), swipe a paginated feed of developers they have not met (Vibe to connect, Ghost to pass); the recipient replies Link or Noped, and a mutual Link unlocks real-time 1:1 chat over Socket.IO with persisted history. Each member gets a shareable portfolio page at their slug (projects, experience, education, certifications, theme, SEO, visibility). httpOnly JWT cookie auth with bcrypt; AWS SES email on every swipe plus a daily 08:00 node-cron reminder for pending Vibes. Loop: Discover to Vibe to Link to Chat to Showcase.

Technologies: React 19, Vite, Redux Toolkit, React Router 7, Tailwind CSS 4, daisyUI, Axios, Node.js, Express 5, MongoDB, Mongoose, Socket.IO, JWT, bcrypt, AWS SES, node-cron, AWS EC2

GlycanBench: a unified resource for working with glycans

Full-stack glycoinformatics platform and installable Python package (glycanbench: SDK, CLI, server) built under Ashok Palaniappan at SASTRA's Systems Computational Biology Lab; live at glycanbench.sastr.edu. Eight tool groups behind one React UI and one FastAPI API: Create (click-to-build glycans with SNFG 2D + 3D conformers via ETKDGv3/MMFF94s, Cytoscape.js biosynthetic networks, IUPAC/WURCS/GlycoCT/SMILES conversion), Visualize (2D SNFG, 3Dmol.js, KEGG pathways), Analyse (characterisation, GlyTouCan insight, RDKit descriptors + fingerprints, motif mutation), Compare (Tanimoto over 5 fingerprints), Align (Needleman-Wunsch with GLYSUM), Cluster (agglomerative/K-means, elbow, outliers), Predict (PyTorch Geometric MPNN immunogenicity classifier, 1,356-glycan dataset, confidence + motif flags), Chat (GlycomicsChat on Groq gpt-oss-120b with an LLM tool router over PubMed, ArXiv, GlyTouCan). 20 routes, ~30 endpoints.

Technologies: Python, FastAPI, Pydantic, React 19, TypeScript, Vite, Tailwind CSS 4, PyTorch, PyTorch Geometric, RDKit, glycowork, glypy, BioPython, scikit-learn, SciPy, LangChain, Groq, 3Dmol.js, Cytoscape.js, Chart.js

EDUCATION

Master of Technology in Big Data Biology

SASTRA Deemed University | Aug 2023 - Jun 2025 | 79.67%

Bachelor of Technology

Prathyusha Engineering College | Aug 2019 - Apr 2023 | 84.4%

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, JavaScript, TypeScript

Data Science & ML: Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, SHAP

Frontend Development: React.js, Next.js, Redux, HTML5, CSS3, Tailwind CSS, Bootstrap

Backend Development: Node.js, Express.js, FastAPI, Flask, Streamlit, Socket.IO, BullMQ, Mongoose

Databases: MongoDB, PostgreSQL, DynamoDB, Oracle, Redis

Cloud & Devops: AWS EC2, AWS SES, Google Cloud, Firebase, Docker, PM2, systemd, nginx, Render, Railway

AI & LLMs: OpenAI GPT-4o, LangChain, LangGraph, Groq, Agentic AI

Bioinformatics & Research: RDKit, Mordred, glycowork, PyTorch Geometric, BioPython, Glycomics, Cytoscape, Galaxy

Tools: Git, GitHub, n8n, Postman, Playwright, Jupyter Notebook, Linux, Vim, Arduino, Photoshop, macOS

CERTIFICATIONS

Namaste ReactJS | NamasteDev

Namaste Node.js | NamasteDev